Summary in Graph

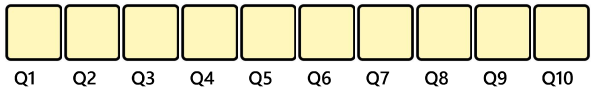
Exam Summary (GO Classes GATE CS | Proof Techniques| Weekly Quiz)

Qs. Attempted:	**0** (0 + 0)	**Correct Marks:**	**0** (0 + 0)
Correct Attempts:	**0** (0 + 0)	**Penalty Marks:**	**0** (0 + 0)
Incorrect Attempts:	**0** (0 + 0)	**Resultant Marks:**	**0** (0 + 0)

Total Questions:	**10** (5 + 5)
Total Marks:	**15** (5 + 10)
Exam Duration:	**30 Minutes**
Time Taken:	**0 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (124) out of 1238 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Consider the following statements:

- S1 : Suppose $a, b, c \in \mathbb{Z}$. If $a^2 \mid b$ and $b^3 \mid c$, then $a^6 \mid c$.
- S2 : Suppose $a \in \mathbb{Z}$. If a^3 is not divisible by 8, then a is odd.

Which of the above statements is true?

A. Only $\mathrm{S1}$.
B. Only $\mathrm{S2}$.
C. Both $\mathrm{S1}$ & $\mathrm{S2}$.
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Consider the following statements:

- $\mathrm{S1}$: If $a, b \in Z$ and $a+b$ is even, then a^2+b^2 is even.
- $\mathrm{S2}$: If $a, b \in Z$ and $a+b$ is odd, then a^2+b^2 is odd.

Which of the above statements is true?

A. Only $\mathrm{S1}$
B. Only $\mathrm{S2}$
C. Both $\mathrm{S1}$ & $\mathrm{S2}$
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Consider the following statements:

- $\mathrm{S1}$: Suppose $a, b \in \mathbb{Z}$. If $\left(a^2\right)\left(b^2-2b\right)$ is odd, then both a and b are odd.
- $\mathrm{S2}$: Suppose $a, b \in \mathbb{Z}$. If $25 \nmid ab$, then $5 \nmid a$ or $5 \nmid b$.

Which of the above statements is true?

A. Only $\mathrm{S1}$
B. Only $\mathrm{S2}$
C. Both $\mathrm{S1}$ & $\mathrm{S2}$
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Consider the following statements:

- $\mathrm{S1}$: If an integer ' a ' is even then $\left(a^2\right)+2a+9$ is odd.
- $\mathrm{S2}$: For an integer ' a ', if $\left(a^2\right)+2a+9$ is odd then ' a ' is even.

Which of the above statements is true?

A. Only $\mathrm{S1}$
B. Only $\mathrm{S2}$
C. Both $\mathrm{S1}$ & $\mathrm{S2}$
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Consider the following statements:

- S1 : Given any $x \in \mathbb{R}$, there exists an element $y \in \mathbb{R}$ for which $xy = 1$.
- S2 : There exists two irrational numbers x and y such that $x + y$ is rational.

Which of the above statements is true?

A. Only S1
B. Only S2
C. Both S1 & S2
D. None

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 2 | Penalty: 0 | Quantitative Aptitude

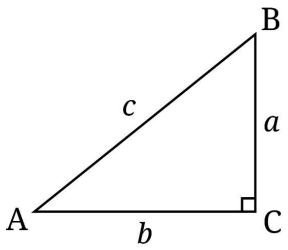
Which of the following statements is/are True?

A. If x and y are two integers whose product is odd, then both must be odd.
B. If a and b are real numbers such that the product ab is an irrational number, then either a or b must be an irrational number.
C. For any integer m, if m^2 is even, then m is even.
D. The sum of a rational number and an irrational number is irrational.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 2 | Penalty: 0 | Quantitative Aptitude

Consider the following right-angled triangle, in which the angle between the side AC and CB is a right angle. Let a, b, c be the length of the sides BC, AC, AB respectively. Assume a, b, c are integers.



Which of the following is/are correct statements?

A. (abc) is even.
B. (bc) is even.
C. (ab) or (ac) is even.
D. Both a, b are even.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 2 | Penalty: 0 | Quantitative Aptitude

Let n be a positive integer.

Which of the following statements is/are true?

A. If m, a are positive integers and $m \mid a^n$, then $m^n \mid a^n$.
B. If p is a prime and a is a positive integer and $p \mid a^n$, then $p^n \mid a^n$.
C. If t is odd, then t^n is also odd, for all n.
D. If t is even, then t^n is also even, for all n.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Consider the following statements:

1. If $n \in \mathbb{Z}$, then $5n^2 + 3n + 7$ is odd.
2. If $n \in \mathbb{Z}$, then $n^2 + 3n + 4$ is even.

Which of the above statements is true?
(Hint: n can be either even or odd)

A. Only 1
B. Only 2
C. Both 1 & 2
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 03sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Quantitative Aptitude

Two integers have the same parity if they are both even or they are both odd. Otherwise, they have opposite parity. Which of the following statements is/are true?

A. If two integers have the same parity, their sum is odd.
B. If two integers have opposite parity, then their sum is odd.
C. If n is an odd integer, then $8 \mid n^2 - 1$.
D. Suppose $a, b \in \mathbb{Z}$. If ab is odd, then $a^2 + b^2$ is even.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

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